
ECON 399 · Introductory Econometrics

Course Outline · Fall 2026

Instructor: Sebastian Fossati
Office: Tory 7-11
Email: sfossati@ualberta.ca

Lectures

- Section A01: Tuesday and Thursday, 11:00 am to 12:20 pm, in T 1-113 (Tory Building).
- Section A03: Tuesday and Thursday, 2:00 pm to 3:20 pm, in T 1-104 (Tory Building).

Remarks:

- There may be classes on which I am unavailable for our *face-to-face* lectures. On those days, the sessions will be on Zoom or pre-recorded.

Office Hours

TBD (posted on Canvas). Office hours may be adjusted throughout the term, and additional hours may be added before exams. On some occasions, office hours will be held on Zoom.

Canvas

I will be using Canvas to post course information and documents. These will include lecture slides, lab handouts, homework assignments and answer keys, the formula sheet, and sample exam questions (without answer keys). Canvas will also be used for homework submission and course announcements.

Course Description

Econometrics uses statistical methods and economic data to answer empirical questions in economics. We are interested in three kinds of questions. (1) How do we test an economic hypothesis? (Example: are men paid more than women with equivalent levels of education?) (2) How do we measure a causal effect? (Example: how much does one more year of schooling raise wages?) (3) What are good methods of forecasting? (Example: what will the inflation rate be next year?)

Questions (1) and (2) are the focus of ECON 399 (and ECON 497). Distinguishing causation from correlation is the central challenge of econometrics. Question (3) is the subject of ECON 409/493 — we cover prediction with regression models, but not time-series forecasting or machine learning.

Learning Goals

The main tool we learn is multiple regression analysis. By the end of the course, students should be able to: (1) specify, estimate, and interpret regression models, and test hypotheses about their parameters; (2) judge when a regression estimate supports a causal claim, and diagnose the common problems — omitted variable bias and other sources of endogeneity, heteroskedasticity, etc. — that arise in practice; and (3) use R to carry out an empirical analysis from raw data to final results.

Course Prerequisites and Corequisites

Prerequisites: ECON 281, 282 and 299 or equivalent, and MATH 156 or equivalent. These pre/corequisites are enforced by the department. If you do not have these pre/corequisites your registration may be cancelled.

The course uses basic probability and statistics and elementary calculus throughout; Appendices A–C of Wooldridge cover the expected background.

Textbooks

Wooldridge (2019, 7th edition) will be our main reference. The course follows this textbook closely, and all chapter references in this outline are to the 7th edition (older editions can also be used).

→ Wooldridge, J.M. (2019). *Introductory Econometrics: A Modern Approach*, 7th edition. Cengage.

In addition, two R companions are available free online (neither is a substitute for Wooldridge):

→ Heiss, F. (2020). *Using R for Introductory Econometrics*, 2nd edition.

Available here: <http://www.urfe.net> (follows Wooldridge chapter by chapter).

→ Hanck, C., Arnold, M., Gerber, A., and Schmelzer, M. (2021). *Introduction to Econometrics with R*.

Available here: <https://www.econometrics-with-r.org>.

Econometrics Package

R is used extensively in the course. R is a free software environment for statistical computing and graphics (<http://www.r-project.org>), typically used through RStudio (also free). No prior experience with R is assumed. We will use the packages `wooldridge` (all textbook data sets), `AER`, and `ggplot2`; each handout also ships with its data as CSV files.

Lab Sessions

As part of the course you will learn how to use R. Weekly lab sessions provide short guided exercises introducing the R skills needed for the homework assignments. Lab attendance is highly encouraged but not graded. The lab schedule will be posted on Canvas.

Evaluation

KEY DATES

- **Midterm 1** in class, Thursday October 1.
- **Midterm 2** in class, Thursday November 5, before fall reading week.
- **Final exam** in the December exam period (December 9–22, 2026).

The final grade will be based on three homework assignments (10% total), two midterm exams (25% each), and a final exam (40%). Further details will be provided as assignments are distributed. No extra credits are available.

Component	Weight	When
Homework ($\times 3$)	10%	one before each exam
Midterm 1	25%	in class, Thursday October 1
Midterm 2	25%	in class, Thursday November 5
Final exam	40%	December exam period (date/time TBD)

Grades reflect judgments of student achievement made by instructors. These judgments are based on a combination of absolute achievement and relative performance in a class.

Remarks:

- Homework assignments will be a combination of computer problems (using R) and analytical problems. The three assignments are equally weighted, with one due before each exam. Everyone must turn in their own assignments, but collaboration is permitted (and encouraged on the computer problems). Late homework assignments will not be accepted. Solutions will follow after the assignments are handed in.
- The midterm exams are short exams held in class. Midterm 1 covers Block 1; Midterm 2 covers Block 2 (with Block 1 material assumed). The final exam is cumulative, with emphasis on Block 3.
- A formula sheet is provided with every exam. Sample exam questions will be made available on Canvas.
- You must not use answers from previous years or answers posted online. If I determine that these conditions were violated, you will receive zero points on the assignment or exam, and the incident will be referred to the Dean's office for further action.

Missed / Deferred Exams

Deferral of term work is a privilege and not a right; there is no guarantee that a deferral will be granted. Misrepresentation of facts to gain a deferral is a serious breach of the *Code of Student Behaviour*.

There will be no make-up midterm exams. A student who misses a midterm exam because of incapacitating illness, severe domestic affliction or other compelling reason (including religious conviction) may have its percentage weight transferred to the final exam. In that case, the student's final exam will be different from that for the rest of the class: it will contain additional questions covering the material of the missed midterm.

A student who misses the final exam because of incapacitating illness, severe domestic affliction or other compelling reason (including religious conviction) may apply for a deferred final exam. Students seeking a deferred exam need to apply to their own Faculty; the instructor does not have the authority to approve such applications.

Laptops and Phones

In class, notes go on paper and screens stay away: **laptops and phones are not permitted during lectures**. Tablets are okay when they lie flat on the desk and are used as a notebook. Students who need a device as part of an *approved* accommodation plan should contact me.

This policy is based on evidence rather than preference. The best study we have on the question is a randomized experiment run in an economics course at West Point, where sections that allowed laptops went on to score lower on the final exam than sections that banned them (Carter et al., 2017). Other research finds that the distraction is not limited to the person using the device: students who can merely see a neighbour's screen also do worse on tests (Sana et al., 2013), and when schools in England banned phones, test scores rose, with the largest gains among the weakest students (Beland and Murphy, 2016). We will discuss this evidence, and what makes it credible, in the course.

Course Material Access program

The Course Material Access program provides undergraduate students with required digital and printed course materials on the first day of class for a flat fee per term that is automatically applied to their account. Students may choose to opt out (or back in) up to one day after the course registration deadline. Visit <https://uab.ca/cmaprogram> to learn more.

The fee is automatically waived for students if **none** of their digital or printed course materials has an associated cost.

Course Outline

The course runs in three blocks. Each block ends with a homework assignment followed by an exam. Chapter references are to Wooldridge (7th edition).

Block 1 • Simple Regression

1. **What is econometrics?** The steps of an empirical analysis; the structure of economic data; causality and the notion of *ceteris paribus*. · Ch. 1
2. **The simple regression model: estimation.** Derivation and properties of the ordinary least squares estimates; goodness of fit; units of measurement and functional form; regression and causality. · Ch. 2
3. **The simple regression model: properties of OLS.** Unbiasedness; sampling variance of the OLS estimators; consistency and large-sample behaviour. · Ch. 2, 5
4. **The simple regression model: inference.** Sampling distributions of the OLS estimators; testing hypotheses; confidence intervals; reporting regression results. · lecture notes

Block 2 • Multiple Regression

5. **Multiple regression analysis: estimation.** Mechanics and interpretation of ordinary least squares; omitted variable bias; efficiency of OLS: the Gauss–Markov theorem. · Ch. 3
6. **Multiple regression analysis: inference.** t tests; confidence intervals; testing multiple restrictions: the F test. · Ch. 4
7. **Multiple regression analysis: further issues.** Data scaling; functional form; goodness of fit and selection of regressors; prediction. · Ch. 6
8. **Qualitative information.** Single and multiple binary (dummy) variables; interactions; a binary dependent variable: the linear probability model. · Ch. 7

Block 3 • Complications

9. **Heteroskedasticity.** Consequences of heteroskedasticity for OLS; heteroskedasticity-robust inference; testing for heteroskedasticity; weighted least squares estimation. · Ch. 8
10. **Serial correlation.** Properties of OLS with serially correlated errors; testing for serial correlation; correcting for serial correlation; robust inference after OLS. · Ch. 12
11. **Endogeneity and causality.** Internal validity; selection bias; instrumental variables; difference-in-differences. · Ch. 15, 13
12. **More on specification and data problems.** Functional form misspecification; outlying observations. · Ch. 9

Student Responsibilities

Generative AI: Generative AI tools (ChatGPT, Claude, Copilot, and similar) are permitted in this course the way a textbook, a classmate, or office hours are: use them to ask questions, review material, and get unstuck — including on R code in the labs and homework assignments. The work you submit must be your own: write up your homework answers yourself, and do not submit AI-generated text, derivations, or code that you cannot explain. AI use is not permitted during exams.

Before outsourcing your homework to AI, consider the design of the course: the homework (10% of the grade) is rehearsal for the exams (90%), which are written in person without AI. Recent large-scale evidence finds that pupils who did their homework with generative AI scored better on the homework but worse on the exams (Strömberg et al., 2026).

Misrepresenting AI-generated work as your own is an academic offence under the University of Alberta *Code of Student Behaviour* and will be treated like any other form of cheating or plagiarism.

Academic Integrity: The University of Alberta is committed to the highest standards of academic integrity and honesty. Students are expected to be familiar with these standards regarding academic honesty and to uphold the policies of the University in this respect. Students are particularly urged to familiarize themselves with the provisions of the Code of Student Behaviour (online at www.governance.ualberta.ca) and avoid any behaviour which could potentially result in suspicions of cheating, plagiarism, misrepresentation of facts and/or participation in an offence. Academic dishonesty is a serious offence and can result in suspension or expulsion from the University.

All students should consult the Academic Integrity website. If you have any questions, ask your instructor.

An instructor or coordinator who is convinced that a student has handed in work that he or she could not possibly reproduce without outside assistance is obliged, out of consideration of fairness to other students, to report the case to the Associate Dean of the Faculty. See the Academic Discipline Process.

Recording of Lectures: Audio or video recording, digital or otherwise, of lectures, labs, seminars or any other teaching environment by students is allowed only with the prior written consent of the instructor or as a part of an approved accommodation plan. Student or instructor content, digital or otherwise, created and/or used within the context of the course is to be used solely for personal study, and is not to be used or distributed for any other purpose without prior written consent from the content author(s).

Recording of Virtual Meetings: If any sessions are held or recorded on Zoom, please note the following disclaimer: “Please note that class times for this course will be recorded. Recordings of this course will be used to allow students enrolled in the course to review the material and will be disclosed to other students enrolled in this section of the class. Students have the right to not participate in the recording and are advised to turn off their cameras and audio prior to recording; they can still participate through text-based chat. It is recommended that students remove all identifiable and personal belongings from the space in which they will be participating.” Recordings will be made available until the end of term and accessible on Canvas. Please direct any questions about this collection to the professor of this course.

Student Resources: The best all-purpose website for student services is: <https://www.ualberta.ca/current-students>.

Accessibility Resources: The University of Alberta is committed to creating work and learning communities that inspire and enable all people to reach their full potential. Accessibility Resources promotes an accessible, inclusive, and universally designed environment. For general information to register for services visit the Accessibility Resources webpage.

The Academic Success Centre: The Academic Success Centre offers a variety of workshops on effective study and exam strategies. There are in-person and online sessions available for a modest fee.

The Centre for Writers: The Centre for Writers offers free one-on-one writing support to students, faculty, and staff. Students can request consultation for a writing project at any stage of development. Instructors can request class visits and presentations.

Learning and Working Environment: The Faculty of Arts is committed to ensuring that all students, faculty and staff are able to work and study in an environment that is safe and free from discrimination and harassment. It does not tolerate behaviour that undermines that environment.

The University of Alberta acknowledges that we are located on Treaty 6 territory, and respects the histories, languages, and cultures of the First Nations, Métis, Inuit, and all First Peoples of Canada, whose presence continues to enrich our vibrant community.

Disclaimer: Any typographical errors in this syllabus are subject to change and will be announced in class and posted on Canvas. The date of the final examination is set by the Registrar and takes precedence over the final examination date reported in this syllabus.

Policy about course outlines can be found in the Evaluation Procedures and Grading System section of the University Calendar.

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